

A TRIZ-Based Method for New Service Design

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This article demonstrates the viability of applying the theory of inventive problem solving (TRIZ) to services by proposing a new approach to new service design. Traditionally, the effectiveness of new service design is unpredictable as service design relies largely on inspiration and the past experiences of service designers. By integrating TRIZ problem-solving tools and its knowledge base, the authors propose a new TRIZ-based approach to address this weakness in service design. Through two case studies, the proposed model is verified. This demonstrates the relevance of TRIZ to service design. It is hoped that this article will raise awareness among service researchers so that more studies in this direction are conducted.

Keywords: service design; TRIZ; new service development; systematic innovation

In the past two decades, the trend of formalizing the process of service development has led to a series of new service development (NSD) models. Although many NSD models comprise a set of development steps starting from product initialization to final commercialization, they are usually oversimplified and lack detailed guidelines for operational use. In particular, recent studies have found that the idea generation stage, despite its importance to the overall service development process, has not been addressed adequately by researchers (Bowers 1989; Edgett 1996; Kelly and Storey 2000). Hence, service developers often have had to use ad hoc processes (Metters, King-

Metters, and Pullman 2003) and tended to not engage in formal ways of idea generation (Easingwood 1986). One potential pitfall of such practices is that the novelty and the quality of new service ideas are severely affected. This is due partly to psychological inertia in human thinking, which focuses the mind on only what is known, and avoiding unknown paths (Savransky 2000). Indeed, service design has been identified as an important research area where more studies should be conducted on the operational tools employed for successful NSD (Menor, Tatikonda, and Sampson 2002). Similarly, Johnston (1999) suggests that good design tools and techniques in service development need to be further explored.

Building on Zhang, Chai, and Tan (2005), this article demonstrates the viability of applying the theory of inventive problem solving (TRIZ) to services by proposing a new method to identify, generate, and evaluate possible solutions to service problems. The outcome of the model is a series of possible solutions that can be further developed into service concepts (Clark, Johnston, and Shulver 2000; Edvardsson and Olsson 1996; Johnson et al. 2001). The article is arranged in the following manner. We first review previous research in service design and the relevant tools. This is followed by a description of the theoretical framework of TRIZ methods. Two empirical case studies are used to illustrate the application of TRIZ in service design. Finally, we discuss the contributions and managerial implications of the study. Based on the limitations identified, several directions for future research are suggested.

LITERATURE REVIEW

According to Johnson et al. (2000), NSD comprises four major phases: design, analysis, development, and full launch. Although the NSD process cycle might represent a progression of planning, analysis, and activity execution, the process is iterative and nonlinear in nature (Menor, Tatikonda, and Sampson 2002). Each phase revolves around the design and configuration of service concept elements where resources such as development teams and tools play a key function. For example, the design stage involves formulation of new services objectives, strategies, idea generation, screening, and concept development and testing. As one of the essential components in the NSD process, service design focuses on the operational basics of the development work. It involves understanding and planning the interaction of a variety of physical, electronic, and human elements (Edvardsson, Thomasson, and Øvretveit 1994). Other definitions include “the idea to design high quality into the service system from the outset, and to consider and respond to customer expectations in designing each element of the service” (Zeithaml, Berry, and Parasuraman 1990, p. 157), the concretization of the service concept in drawing flowcharts (Gummesson 1991), and “the work of specifying an idea about a new service in drawings and specifications” (Norling, Edvardsson, and Gummesson 1992, p. 57). Previous work in service design has identified the need to incorporate more customer elements throughout the process of NSD. One good example of this is the development of an inventory of activities that needs to be carried out in involving users in NSD projects (Alam 2002). Another example is the use of consumer choice modeling in e-services, which provides insights in understanding consumer choices and preferences for design and operation strategy formulation in NSD (Iqbal, Verma, and Baran 2003).

Among the many activities in service design, idea generation is one of the most critical activities. Ideas for new product can arise almost anywhere (Robinson and Moak 1997). They can arise inside the company by taking employees as an important source of new product ideas (McGuire 1973), or they can come from outside the company by listening to customers’ feedback or copying from competitors (Easingwood 1986; Sheuing and Johnson 1989). Conventional idea search methods include brainstorming, solicitation of ideas from employees and customers, lead-user research, and learning about competitors (Zeithaml and Bitner 2000). However, these methods rely very much on service designers’ or customers’ past experiences, which contain unavoidable psychological inertia. Thus, the adoption of these methods might probably restrict the generation of creative and breakthrough ideas.

Although idea generation is widely agreed to be important, it is one of the weakest links in NSD (Bowers 1989). Many service companies tend not to engage in formal idea generation in that they find it relatively easy to generate new service ideas (Easingwood 1986). However, recent studies report that successful service firms establish systems and procedures for stimulating idea generation on a long-term basis (Edgett 1996; Felberg and DeMarco 1992; Robinson and Moak 1997). Techniques for stimulating ideas must be created and maintained on an ongoing basis to keep generating innovations (Crawford 1994).

Another focus of research in service design is the development of tools and techniques. Service design tools such as service blueprinting (Baum 1989; George and Gibson 1991; Kingman-Brundage 1988; Shostack 1984), functional analysis (Berkley 1996), structured analysis and design technique (Congram and Epelman 1995), quality function deployment (QFD) (Ermer and Kniper 1998; Franceschini and Terzago 1998; Selen and Schepers 2001; Stamm 1992), and root cause analysis (RCA; Duffy 2000) have been developed to assist service designers in developing design concepts for services. Although these service design tools may be effective in describing and analyzing service problems, they have limited capacity in generating ideas and overcoming pitfalls identified in the design process. This limitation in overcoming psychological inertia affects both the quantity and the quality of subsequent design solutions. The challenge is compounded by the ill-structured nature of design problems, in which often one or more steps (or states) are either unknown or incoherent, there is insufficient information in the initial state, and the properties of the goal state are never fully specifiable in advance (Goldschmidt 1997). For example, as part of QFD analysis, the house of quality (HoQ) is frequently used by service researchers to identify problems, such as unveiling contradictory relationships among different service product attributes (Rovira and Aguayo 1998). However, apart from problem identification, it is noted that HoQ often fails to provide effective solutions to eliminate the contradictions without making compromises between conflicting requirements. Another example is RCA, which is often used in identifying potential service failure points. The preventive solutions generated are based mainly on past experiences of designers rather than as the result of RCA. Moreover, because RCA is closely allied to the optimization of existing processes (Mann 2002), if service designers rely wholly on the outcome of its analysis, it would be very difficult to find novel breakthrough solutions.

SERVICE DESIGN BASED ON TRIZ

To overcome the limitations in service design tools and idea generation practices, TRIZ, a systematic problem-solving methodology, is proposed in this study to close the gaps. TRIZ was first developed by Genrich Altshuller and his colleagues in the former USSR, starting in 1946. The hypothesis of TRIZ research is that there are universal principles of invention that are the basis for creative innovation. If these principles could be identified and codified, they could be taught to people to make the process of invention more predictable. Through the analysis of more than 2 million patents, a number of innovation patterns and laws of ideality were identified. In addition, the work also reveals the following:

- Problems and solutions were repeated across industries and sciences.
- Patterns of technical evolution were repeated across industries and sciences.
- Innovations used scientific effects outside the field where they were developed.

Therefore, TRIZ can be described as a structured problem-solving process with the integration of a set of problem definition and resolution tools that was created on the basis of patent analysis. A distinct characteristic of TRIZ problem solving is systematic problem resolution without compromise. The entire problem-solving process is guided by TRIZ tools, which direct the problem solver to explore solutions in directions that have previously been proven successful. A typical TRIZ problem-solving process comprises three stages: problem definition, problem resolution, and solution evaluation. At each stage, a variety of TRIZ tools can be used. For instance, the Innovative Situation Questionnaire, the ideal final result, and function analysis are frequently used in problem definition. The 40 inventive principles, 4 separation principles, the algorithm of inventive problem solving (ARIZ), and 76 standard solutions are often used in problem resolution.

The Relevance of TRIZ to Services

According to TRIZ methodology, most problems have so-called inherent contradictions. A typical contradiction is a conflict due to two conflicting requirements to the same element in a system. It could result also from two conflicting elements in the same system. The identification of an inherent contradiction is a critical step in TRIZ problem analysis. Once a contradiction is formulated, selected TRIZ knowledge-based tools can be used to eliminate the contradiction. The resolution of the contradiction usually leads to the resolution of the main problem as well as other minor problems.

Not surprisingly, contradictions in technical areas are more tangible and easier to formulate. These contradictions are often clear-cut conflicts between two design parameters. These have been summarized in the original TRIZ contradiction matrix (Altshuller 1997). However, it is possible to formulate problems in services in the form of contradiction despite its intangible nature. For example, "mass customization" is a typical service contradiction with implications in many service sectors. How can a service firm deliver personalized service offerings, based on individual customer requirements, to a large group of customers without incurring additional costs brought about by the personalization? Is there any breakthrough solution that could lead to a win-win situation without compromising either "standardization" (to achieve economy of scale) or "customization"? Because TRIZ has the ability to eliminate contradictions and generate breakthrough solutions, it is possible to use TRIZ to find creative win-win solutions. To explore this opportunity, a revised version of the 40 inventive principles (see Appendix A, adapted from Zhang, Chai, and Tan 2003) was developed with the aim of applying them in resolving service contradictions.

The other positive overlap between TRIZ and services is the research on the patterns of innovations. Through the analysis of many examples, Berry and Lampo (2000) categorized five typical ways of redesigning service offerings (i.e., self-service, direct service, preservice, bundled service, and physical service). This suggests that the patterns of service innovations can be predicted. This assumption is similar to the underlying philosophy of the TRIZ 40 inventive principles. The comparison shown in Table 1 indicates many similarities between these five service redesign patterns and the 40 inventive principles. One implication of this comparison is that if service designers understand the patterns of innovations that TRIZ has successfully demonstrated for physical products, the process of service design and innovations can be made more predictable.

Service Design Based on TRIZ

TRIZ can help to eliminate the psychological inertia in the minds of service designers, thus enhancing the capacity of service idea generation in service design. The knowledge-based toolkit provided by TRIZ caters exactly to the needs of helping problem solvers to overcome their own psychological inertia, which is often considered the hardest part in solving difficult problems (Altshuller 1984). Mann and Dewulf (2002) also argued that in terms of its toolkit and method, TRIZ is the most comprehensive of all available models. Compared to traditional idea generation methods such as brainstorming, synectics, lateral

TABLE 1
Similarities Between Service Redesign Patterns and TRIZ 40 Inventive Principles

<i>Service Redesign Pattern</i>	<i>Match of 40 Inventive Principles</i>
Self-service: Customer assumes role of producer	Principle 25: Self-service. Make an object serve itself by performing auxiliary helpful functions. Use waste resources, energy, or substances
Direct service: Service delivered to the customer's location	Principle 2: Taking out. Separate an interfering part or property from an object or single out the only necessary part (or property) of an object.
Preservice: Streamlines activation of the service	Principle 10: Preliminary action. Perform, before it is needed, the required change of an object. Prearrange objects such that they can come into action from the most convenient place and without losing time for their delivery.
Bundled service: Combines multiple services into a package	Principle 5: Merging. Bring closer together (or merge) identical or similar objects, and assemble identical or similar parts to perform parallel operations. Make operations contiguous or parallel; bring them together in time.
Physical service: Manipulation of tangibles associated with the service	Principle 15: Dynamics. Allow (or design) the characteristics of an object, external environment, or process to change to be optimal or find an optimal operating condition.

NOTE: TRIZ = theory of inventive problem solving.

thinking, morphological analysis, and mind mapping, Savransky (2000) argued that only TRIZ is useful for solving most difficult problems, which are classified into the type that has unknown causes and unknown search directions. Because those traditional methods are frequently used in service idea generation (Edvardsson et al. 2000), it is also likely that TRIZ can be applied to and benefits service design.

However, a search on previous studies applying TRIZ in the service domain reveals that little work has been done, with the exception of some preliminary works by Low et al. (2001) and Rantanen and Domb (2002), who briefly mentioned the possibility of applying TRIZ to services. To explore the viability of applying TRIZ to services, we developed a method for new service design. The following section describes this approach and its application.

Theoretical Framework

Based on the principles of TRIZ, we propose a new approach to systematic service design that comprises three main stages (see Figure 1). The input to the method is an identified service problem. At Stage 1, the identified problem is "translated" into the language of TRIZ to provide insightful information for further problem solving. At Stage 2, the problem is structured into typical TRIZ contradictions by conducting contradiction analysis, one of the most effective problem resolution tools. TRIZ knowledge-based principles such as 40 inventive principles and 4 separation principles may then be employed to eliminate the contradictions. At Stage 3, the generated ideas are evaluated by using the unique TRIZ criteria, which is the ideal final result. The final output is a list of possible innovative conceptual solutions to service design. If solutions are still not found after contradiction elimina-

tion, or some other new problems occur after solution evaluation, the problem-solving process will be iterated back to the first stage to redefine the original situation. The next few paragraphs describe the three stages in more detail.

Stage 1: Problem Definition

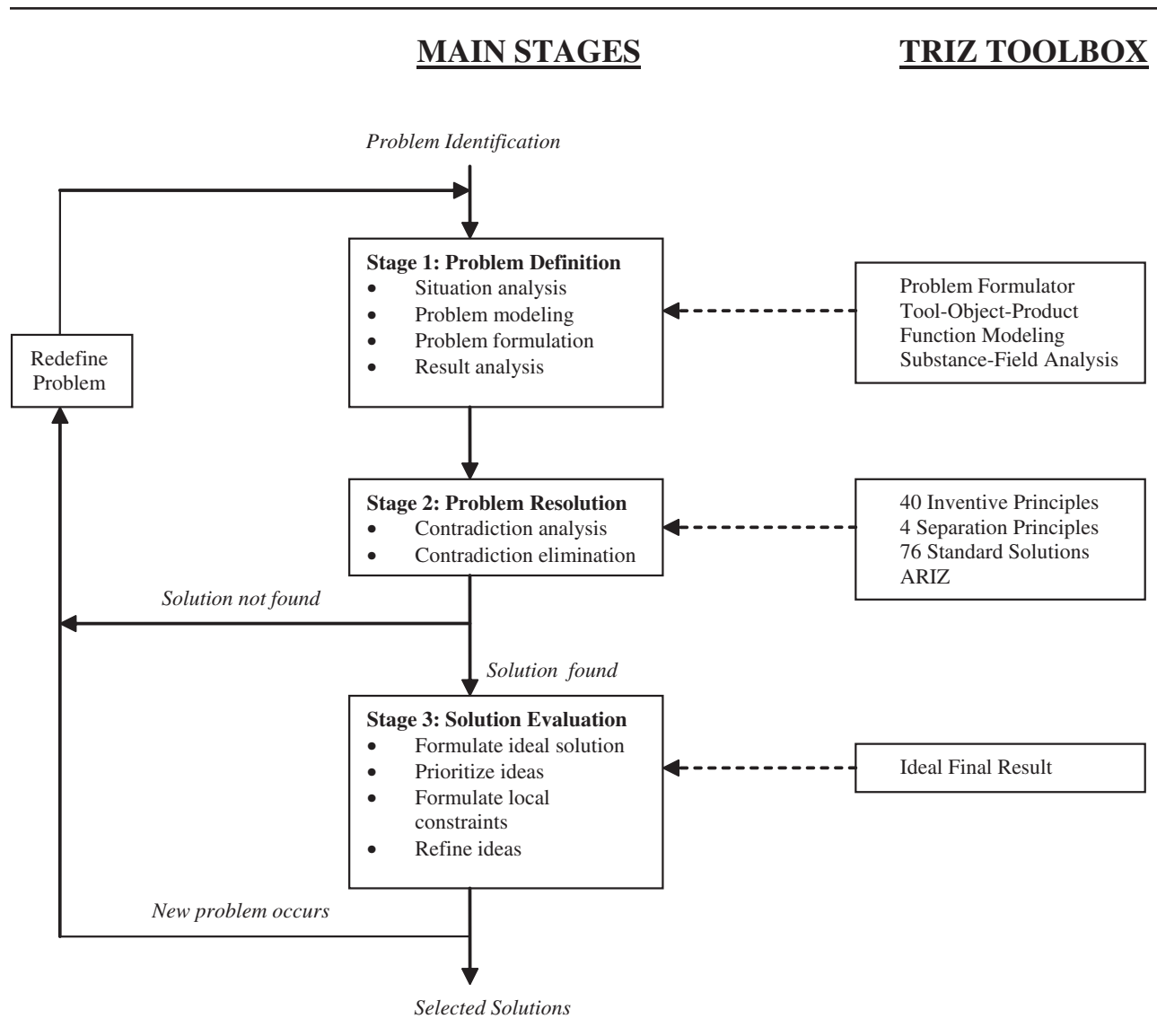
The objective of problem definition is to translate problem information into the language of TRIZ for the preparation of problem resolution. This stage consists of four steps: (a) situation analysis, (b) problem modeling, (c) problem formulation, and (d) result analysis.

Situation analysis. Situation analysis is used to conduct preliminary analysis and elicit useful information about the problem situation. Based on the Innovative Situation Questionnaire (Terninko, Zusman, and Zlotin 1998) of TRIZ, we developed a set of questions (see Table 2) to help service designers to better understand the in-depth situation of the original problem. Detailed answers to the questions will be used to stimulate the generation of innovative solutions at the next stage of problem resolution.

Problem modeling and formulation. After obtaining information from the situation analysis, the next step is problem modeling and formulation. Problem modeling refers to building a function diagram by using function analysis. Problem formulation is about developing an exhaustive set of problem statements on the basis of the function diagram. These can be achieved by using TRIZ tools such as the problem formulator (Zlotin and Zusman 2001), substance-field analysis (Terninko, Zusman, and Zlotin 1998), and tool-object-product function modeling (Royzen 1999).

Take the problem formulator, for example. The process of building a function diagram begins with the identifi-

FIGURE 1
Problem-Solving Model for New Service Design Based on TRIZ



cation of the basic functional components. This is followed by specifying the relationships among the functions. Function identification can be done by asking what the service does for consumers (Berkley 1996). Problem formulation classifies system functions into two types: harmful functions and useful functions. The identified functions are connected to each other in the form of a network of cause-and-effect relationships.

Next, two types of problem statements can be formulated based on the relationships of function components. They are preventive statements for harmful functions and

alternative statements for useful functions. An additional statement for each node can be expressed as a benefit from a harmful function, an enhancement to a useful function, or the resolution of a contradiction (Terninko, Zusman, and Zlotin 1998).

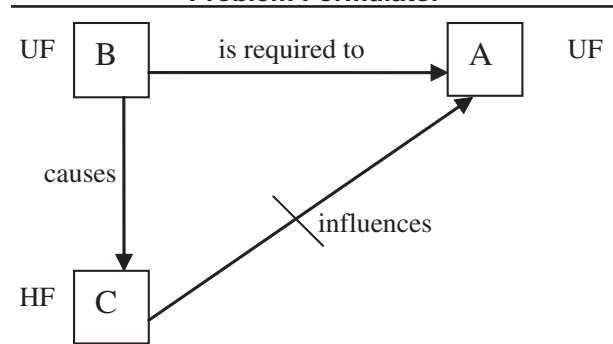
Figure 2 is a simple example that illustrates the use of the problem formulator in problem modeling and formulation. Suppose that through situation analysis, three functions (two useful functions, A and B, and one harmful function, C) are identified in the original system. Function B is the prerequisite to deliver Function A. However, the

TABLE 2
Questionnaire for Situation Analysis

1. What is the purpose of the target service?
Describe what are the (potential) customer needs to be met and how existing service, or desired new service, aims to meet the needs.
2. What are the existing problems?
Describe briefly the existing problems in the service operations system.
3. What are the known solutions?
State the past and current solutions to resolve identified problems, remove barriers, or improve/refine the situations.
4. What are the advantages and disadvantages of the known solutions?
5. What is the structure of the target service operations system?
Identify the subsystems or components of the target service system, and then specify the relationships among subsystems or components.
6. What is the ideal solution to the original problem?
Formulate all ideal solutions that deliver all of the benefits without compromising from any harmful elements, and incur no costs to solve the problem.
7. What are the local constraints or limitations?
Estimate a feasible cost budget for solving the problem, and find out what are acceptable and unacceptable changes to the original system.
8. What is the objective for this problem-solving project?
Formulate the problem-solving objective. A typical objective of TRIZ problem solving is to eliminate harmful elements in the system without introducing new problems and deteriorating the original system.

NOTE: TRIZ = theory of inventive problem solving.

FIGURE 2
Illustrative Example of Using the Problem Formulator



NOTE: UF = useful function; HF = harmful function.

use of Function B will produce harmful Function C, which can have a reverse effect on the effectiveness of Function A. The functional diagram can be built based on the cause-and-effect relationships among the three different functions. Based on the functional diagram, seven problem statements can be formulated as follows:

1. Find an alternative way to obtain A that does not require B and will not be influenced by C.
2. Find a way to enhance the effectiveness of A.

3. Find an alternative way to obtain B that provides A and does not cause C.
4. Find a way to enhance the effectiveness of B.
5. Find a way to resolve the contradiction that B should provide A and B should not cause C.
6. Find a way to eliminate, reduce, or prevent C under the condition of B.
7. Find a way to benefit from C.

Essentially, formulating problem statements decomposes a complex problem into a series of correlated small problems, which makes it easier and more straightforward to solve.

Result analysis. After problem formulation, professional knowledge should be used to solve formulated and decomposed "problem series." Very often, some solutions or at least some indications to possible solutions can be obtained through analyzing the problem statements.

However, sometimes when the function diagram is very complex and consists of many components, there are many problem statements. It would be time-consuming to analyze each problem statement. Terninko, Zusman, and Zlotin (1998) propose a set of criteria to select the most "probable" problem statements. This should direct service designers to resolve problems in an efficient manner. The criteria are as follows:

- Select the problem with the best cost/benefit ratio.

- The more radical the problem, the greater the potential benefit.
- It is better to eliminate a harmful cause than to mitigate results.
- The level of difficulty involved in implementing a solution should be a factor in problem selection. Too radical a solution may prove unacceptable, depending on an organization's culture and psychological inertia.

Stage 2: Problem Resolution

The two major activities in this stage are contradiction analysis and elimination. Effective solutions may emerge after problem definition. However, when situations are complicated, the process of problem modeling might be very time-consuming and would probably result in a lengthy set of problem statements for analysis. An alternative way to analyze problems is to identify one or just a few of key problems (inherent contradictions) that are the causes for many other problems within the same system. The elimination of the inherent contradictions will usually also lead to the resolution of these problems.

Contradiction analysis. The purpose of structuring an inventive problem into the form of a contradiction is to identify two conflicting components (either subsystems or functions) in the original problem systems or two opposite requirements to the same element/condition of the system. Sometimes, contradictions can be found by analyzing problem statements, defining the tool-object-product in the system, or simply by using the TRIZ technique of root contradiction analysis (Mann 2002).

Contradiction elimination. To eliminate formulated contradictions effectively, one may use any knowledge-based tools provided by TRIZ, which include the 40 inventive principles, the contradiction matrix, 76 standard solutions, and the ARIZ. Among them, the 40 inventive principles are considered one of the most accessible and useful TRIZ problem resolution techniques. Practical applications have proved that these principles are not only effective in eliminating contradictions for technical problems (Altshuller 1997), but they are also equally effective in handling nontechnical problems (Mann and Domb 1999; Retseptor 2003; Terninko, Zusman, and Zlotin 2001). Appendix A and Appendix B provide examples to demonstrate the effectiveness of using the 40 inventive principles and 4 separation principles in providing innovative solutions to resolve problems in the service arena. With the use of these principles, service contradictions can be eliminated and a list of effective solutions generated for further assessment in terms of their closeness to ideality.

Stage 3: Solution Evaluation

The last stage of the proposed method involves evaluating the solution using TRIZ's principles. The basis of TRIZ idea evaluation is based on the law of ideality (Altshuller 1997). According to this law, the best idea should be the one closest to the state of ideality. This can be measured by using the technique of the ideal final result. The ideal final result is an implementation-free description of the situation after a problem has been solved (Domb 1997). An ideal solution delivers all of the desirable benefits without harm and incurs no cost to solve the problem. The selected idea to be implemented can be further refined until it is accepted by local constraints. Otherwise, another idea alternative can be promoted for analysis.

EMPIRICAL STUDY

To verify the viability of the proposed problem-solving model, two case studies were conducted at two different service contexts. One was at a leisure resort island. The other was at a university. The case studies demonstrate that the proposed method can be used to help resolve problems in service operations and to generate quality solutions for implementation. They also indicate that this service design may be applicable to resolve problems in different service industries. Table 3 provides a summary of the structure of the two case studies and the linkage to the TRIZ service problem-solving process reflected in Figure 1. It demonstrates how the flow of the TRIZ process links to the resolution of service problems.

Case 1: Redesign the Sightseeing Scheme to New Areas on Singapore Sentosa Island

To attract more visitors to the Singapore Sentosa Island, a leisure resort island in Singapore, the Singapore Development Corporation (SDC) planned to open more previously inaccessible areas of the island to visitors. To make visitors better enjoy traveling to these new areas, the SDC developed a new leisure program where visitors can either bring their own bicycles or rent bicycles from one of three booths on the island. Renting a bicycle costs visitors between \$4 and \$8 an hour. Another reason for developing this program was that it is difficult to visit much of the island on foot. For this purpose, a project of designing and carving out new cycling paths was carried out. However, some problems surfaced after the implementation of this cycling program. How can tourists who do not prefer cycling or cannot cycle at all visit the new areas? Was it

TABLE 3
Case Study Structure and the Linkage to the TRIZ Service Design Model

<i>TRIZ Service Design Stage</i>	<i>TRIZ Tools</i>	<i>Case 1: Redesign Sightseeing Program on Sentosa Resort</i>	<i>Case 2: Restructure University Canteen Operations</i>
Problem definition	Problem formulator (function modeling and problem statement formulation)	The objective is to design new programs to meet the needs of tourists who visit Sentosa Island without the necessity of cycling. The service context of the visiting program and tourists' traveling needs is integrated into the function diagram to reflect the problem situation.	The objective is to restructure canteen operations to meet the dining needs of customers with more flexible opening hours. The service context of canteen operations and customer needs is functionally expressed in a diagram to reflect the problem situations.
Problem resolution	Contradiction analysis; 40 inventive principles and 4 separation principles	A contradiction in travel route design is formulated. Several inventive principles and separation principles are employed to eliminate contradiction and generate solutions.	A contradiction in operation time is formulated. Inventive principles and separation principles are employed to eliminate contradiction and generate solutions.
Solution evaluation	Ideal final result	Generated solutions are sorted out against the feasibility of ideas.	Generated solutions are sorted out against the feasibility of ideas.

NOTE: TRIZ = theory of inventive problem solving.

worthwhile to invest in paving cycling paths? Are there any better means or supplementary methods to help visitors access the new areas? In this case study, the TRIZ service design method was employed to analyze the service context and work out a set of possible strategies to address the problem situation.

STAGE 1: PROBLEM DEFINITION

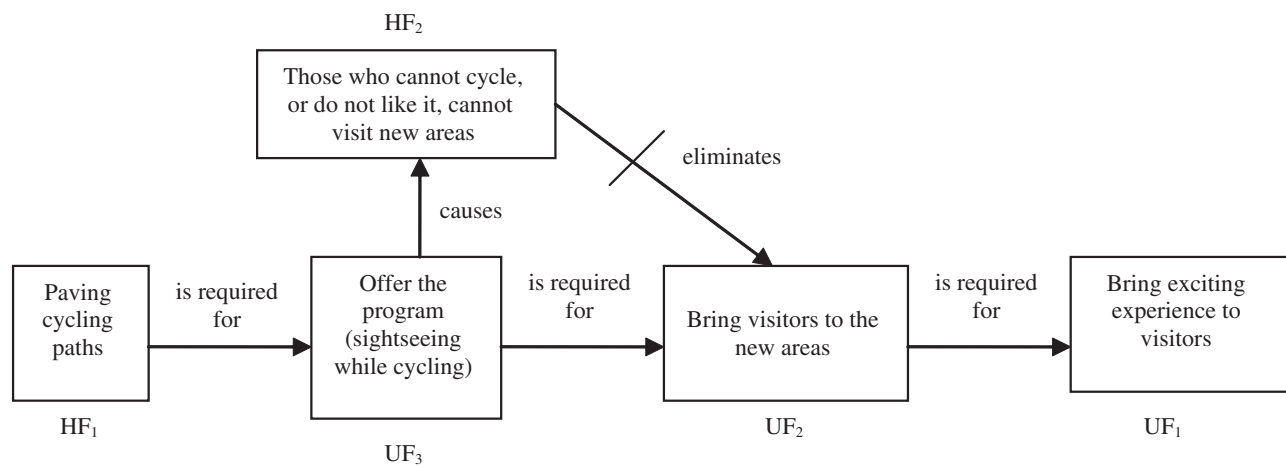
The advantage of the program of sightseeing while cycling might be its flexibility for visitors. Some visitors, especially young ones, might prefer this method. However, the majority of visitors who come to Sentosa Island may not have exercising in mind. Thus, they might not be pleased if they are not able to enjoy the views in the new areas if they do not cycle. Other drawbacks include the added cost to rent the bicycles and the investment by the SDC in building the cycling paths. The operations system of the cycling problem consists of components such as visitors, renting booths, operation staff, bicycles, and cycling paths on the island. The objective for solving this problem could be to formulate effective strategies to redesign the current program or develop a new program that can eliminate the existing drawbacks without introducing new harmful elements to the original service operations system.

Based on the preliminary problem analysis, a function diagram of the target cycling problem is constructed (see Figure 3).

According to the function diagram, the following 11 problem statements can be formulated:

1. Find an alternative way to obtain [UF₂: Bring visitors to the new areas] that provides [UF₁: Bring exciting experience to visitors], does not require [UF₃: Offer sightseeing while cycling program], and is not influenced by [HF₁: Those who cannot cycle, or do not like it, cannot visit new areas].
2. Find a way to enhance the effectiveness of [UF₂: Bring visitors to the new areas].
3. Find an alternative way to obtain [UF₁: Bring exciting experience to visitors] that does not require [UF₂: Bring visitors to the new areas].
4. Find a way to enhance the effectiveness of [UF₁: Bring exciting experience to visitors].
5. Find an alternative way to obtain [UF₃: Offer sightseeing while cycling program] that provides [UF₂: Bring visitors to the new areas], does not cause [HF₁: Those who cannot cycle, or do not like it, cannot visit new areas], and requires [HF₂: Paving cycling paths].
6. Find a way to enhance the effectiveness of [UF₃: Offer sightseeing while cycling program].
7. Find a way to resolve the contradiction that [UF₃: Offer sightseeing while cycling program] should be in place to provide [UF₂: Bring visitors to the new areas] and should not exist to avoid causing [HF₁: Those who cannot cycle, or do not like it, cannot visit new areas].
8. Find a way to eliminate, reduce, or prevent [HF₁: Those who cannot cycle, or do not like it, cannot visit new areas] to avoid hindering [UF₂: Bring visitors to the new areas] under the condition of [UF₃: Offer sightseeing while cycling program].

FIGURE 3
Function Diagram of the Cycling Program on Sentosa Island



NOTE: UF = useful function; HF = harmful function.

9. Find a way to benefit from [HF₁: Those who cannot cycle, or do not like it, cannot visit new areas].
10. Find a way to eliminate, reduce, or prevent [HF₂: Paving cycling paths] under the condition of [UF₃: Offer sightseeing while cycling program].
11. Find a way to benefit from [HF₂: Paving cycling paths].

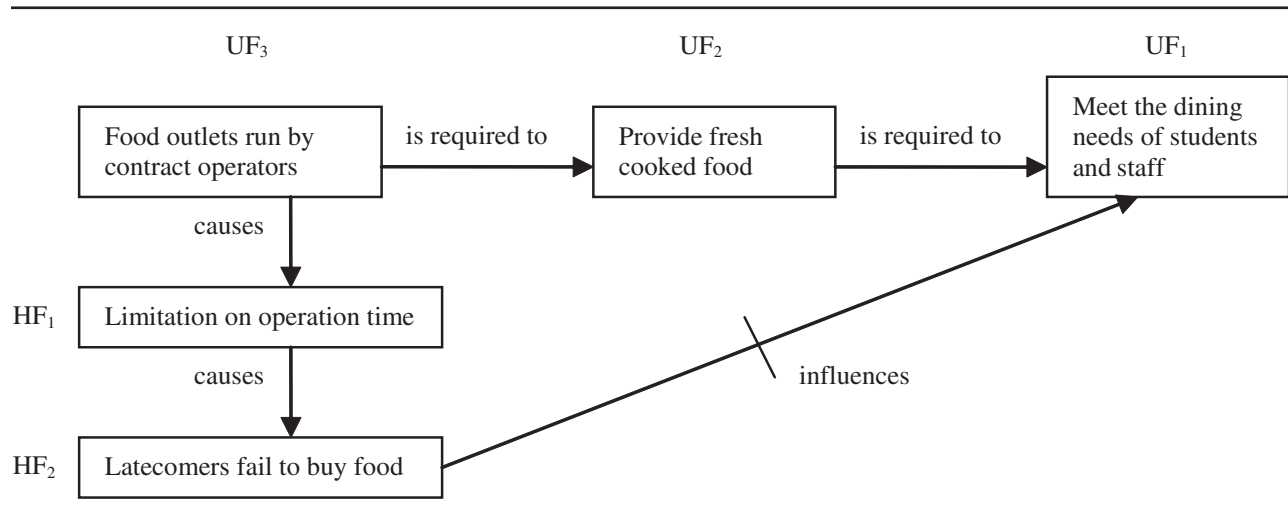
The analysis of these problem statements can give some indications of possible solutions, which may be insightful, whereas others may not be practical. The following are some indications based on the analysis of the formulated 11 problem formulations:

- Find some alternatives to help visitors enjoy the scenery of the new areas but not to visit physically (e.g., tapes, pictures?).
- Design other programs to entertain visitors instead of opening the new areas (e.g., develop more attractions in old areas?).
- Find some alternatives to get visitors around the island instead of sticking to the original program of sightseeing while cycling (e.g., cable car, helicopter, yacht?).
- Find lower cost alternatives to the suggestion of "paving cycling path" (e.g., design an airway or seaway?).

STAGE 2: PROBLEM RESOLUTION

From the perspective of contradiction analysis, the situation can be interpreted as a problem of travel route design, which can be one of the core problems behind all problems. It can be structured as the contradiction that the traveling path should be present to direct the visiting with the "riding vehicle," and yet the "path" should be absent to increase the flexibility of visiting, save costs for path paving, and lower its effect on the natural surroundings. To resolve this contradiction, we can first intensify the contradiction to two extreme situations: The path is well designed and may have more functions than just for the use of cycling. For example, the current path can be widened for electric-powered buses or a tram system that can take visitors to anywhere on the island. The other opposite extreme scenario is that no path be provided, which may be interpreted as looking for some intangible or flexible paths or just encouraging customers to get there without the need of paved paths. Some examples could be offering new means to take visitors to these new areas while adding some fun elements, such as introducing yachting, taking cable car, or even riding on an hot-air balloon. The alternative way to obtain the above strategies is to use a set of inventive principles such as "another dimension," "self-service," and "segmentation," and to use the separation principles such as separation in time and separation in space.

FIGURE 4
Functional Diagram of
University Canteen Food Outlet Operations



NOTE: UF = useful function; HF = harmful function.

STAGE 3: SOLUTION EVALUATION

From the perspective of the ideal final result, if the visitors are keen to join an event such as an excursion or a party that is organized in new areas and find their own ways to get there, this solution might be the closest to ideality. More incentives, such as freebies and rewards, can be designed in a way to motivate the passion of exploration for visitors.

Case 2: Restructure the Operations at a University Canteen

This case study (adapted from Zhang, Chai, and Tan 2005) was carried at a university canteen. The canteen only opens during weekdays to provide breakfast, lunch, and dinner for students and staff of the university. This constraint in operation hours creates problems to the dining needs of many people. Many diligent research students and staff who work on campus beyond office hours or on weekends have to travel far to find a restaurant or food stall. Complaints were made to the university to seek a change in the canteen operating hours.

STAGE 1: PROBLEM DEFINITION

The situation information was captured through observation and a number of interviews with students and staff who dine at the canteen frequently. The operation of the canteen and food outlets was also investigated through dis-

cussions with the canteen staff. Using the situation analysis questions (Table 2), the objective, system, and ideality were identified as follows:

- *Objective.* The purpose of improving this situation is to find some effective measures to improve the food supply in the canteen so that the dining needs of all students and staff can be met.
- *System.* The system of interest in this problem is university-wide. The center of the system is the canteen, which includes all the physical facilities, food outlet operators, and other resources in the canteen. The customers to be served in the canteen are the students and staff.
- *Ideality.* The ideal solution should be free of all the existing problems in the system of the canteen without incurring extra cost.

By using the problem formulator, a set of events is extracted and linked with each other using cause-and-effect relationships (see Figure 4). Based on the functional diagram, 11 problem statements are formulated as follows:

1. Find an alternative way to obtain the useful function of [UF₁: Meet the dining needs of students and staff] that does not require [UF₂: Fresh-cooked food] and is not influenced by [HF₂: Latecomers fail to buy food].
2. Find a way to enhance the effectiveness of [UF₁: Meet the dining needs of students and staff].

3. Find an alternative way to obtain the useful function of [UF₂: Fresh-cooked food] that can provide [UF₁: Meet the dining needs of students and staff] and does not require [UF₃: Food outlets run by contracted operators].
4. Find a way to enhance the effectiveness of [UF₂: Fresh-cooked food].
5. Find an alternative way to obtain the useful function of [UF₃: Food outlets run by contracted operators] that provides [UF₂: Fresh-cooked food] and does not cause [HF₁: Limitation on operation time].
6. Find a way to enhance the effectiveness of [UF₃: Food outlets run by contracted operators].
7. Find a way to resolve the contradiction that [UF₃: Food outlets run by contracted operators] should be established to provide [UF₂: Fresh-cooked food], but it should not be established to avoid causing [HF₁: Limitation on operation time].
8. Find a way to eliminate, reduce, or prevent the harmful function of [HF₁: Limitation on operation time] to avoid causing [HF₂: Latecomers fail to buy food] under the condition of [UF₃: Food outlets run by contracted operators].
9. Find a way to benefit from [HF₁: Limitation on operation time].
10. Find a way to eliminate, reduce, or prevent the harmful function of [HF₂: Latecomers fail to buy food] to avoid influencing [UF₁: Meet the dining needs of students and staff] under the condition of [HF₁: Limitation on operation time].
11. Find a way to benefit from [HF₂: Latecomers fail to buy food].

An analysis of the above statements can guide thinking toward the following directions, through which some possible solutions can be interpreted:

- Find alternative ways of delivering food, or replacements of fresh-cooked food, so that people can come and buy at anytime (e.g., direct sale of ordered food, automated vendor machine, canned food; provide office pantry so that students can cook or heat food).
- Find alternative ways to obtain fresh-cooked food without relying on contracted operators (e.g., contract with off-campus operators who can operate without the time constraint).
- Find alternative ways to operate existing food outlets without the time constraints (e.g., encourage a few existing operators to extend operation time).
- Find alternative ways to serve customers without the time constraint (e.g., new packaging to sustain the freshness of food so that operators can cook food on order and then deliver it to designated places where customers can collect food).
- Find some means to squeeze the customer demands cycle (e.g., early bird discount).

STAGE 2: PROBLEM RESOLUTION

It is not difficult to identify the two conflicting aspects in the original system: the operation time of food outlets and customer demands. The contradiction can be structured such that operation time should be long enough to meet the dining needs of students and staff. Yet opening hours should not be too long because it is not cost-effective to the food outlet operators. The essence of eliminating this contradiction is to take effective measures to either stretch the operation time or to concentrate the demands of the customers. Based on the extreme situation analysis, the 4 separation principles, and the 40 inventive principles, we can find out some useful results to eliminate the contradiction and also other supplementary measures by interpreting some of the principles in the problem context. It is interesting to note that some insightful indications that have been obtained at Stage 2 can also be found here.

- *Separation in space.* Separate the food preparation from food supply by contracting food preparation to off-campus operators and use phone ordering, direct delivery, or other means to supply food in batches to campus.
- *Separation within a whole and its parts and segmentation.* Segment the dining needs of customers into different types and patterns. Categorize those who have particular needs such as late dining and provide special service to them.
- *Separation in time.* Divide the operation of the canteen food outlets into daytime operation and nighttime operation so that other operators can use the existing outlets to operate at night.
- *Self-service.* In the deliver-on-order service, customers can collect the ordered food by themselves at designated collecting points. Pantries can be provided in offices so that late diners can self-resolve their problems by cooking or heating their own food.
- *Preliminary action.* Set up some complementary measures such as providing automated food vendor machines and renovating pantries to relieve peak-hour demand and also to meet the needs of latecomers.

STAGE 3: SOLUTION EVALUATION

The ideal solution to address this problem should incur as little cost as possible while providing maximum benefit to the system of canteen operations. Among the obtained ideas from the above problem-solving steps, the simplest solution is to identify one or two outlet operators who are willing to extend their operation time. Another possibility

is to introduce new operators to take over the business of night operation. To implement this idea, support must be obtained from the university to give an incentive for these operators who serve students and staff at night. Another approach to address this problem is to launch a food-ordering service, which can help students and staff to order their food over the telephone or through the campus Internet. The food can be prepared by contract operators, who can be either from the existing canteens or from off-campus restaurants. The ordered food can be delivered to designated collection points or delivered directly to offices, and a service fee is charged.

DISCUSSION

Contribution of TRIZ Service Design Method

TRIZ is a unique way of systematic thinking. With a powerful knowledge base as its foundation, TRIZ contrasts with other problem-solving approaches by its distinct capabilities to generate breakthrough ideas and deliver solutions without compromise. New service design can benefit from this breakthrough problem-solving method. Unlike many existing service design tools, this TRIZ-based approach can help service developers to overcome their psychological inertia and generate many new and even breakthrough ideas throughout the process of service conceptualization. For instance, the effectiveness of QFD in service design relies very much on the past experiences of participants involved in the service design exercise. Brainstorming is required to provide insight into the understanding of service product attributes and the scoring of the satisfaction level for these attributes. In contrast, the main concerns for TRIZ problem solving are service context and service activities, which are much less experience dependent. In terms of quality and efficiency of TRIZ problem solving in service design, it is possible that under some circumstances, an experienced service designer might probably develop better and faster service solutions without using a systematic approach. However, this is exactly the inherent weakness of current practices in NSD and many existing service design methods that rely heavily on previous specific knowledge and experience. By using TRIZ, one can always rely on its knowledge base even when service developers and participants are not experienced. This is the novelty of using TRIZ in NSD, which is also the biggest difference between TRIZ and other existing service design methods.

In this article, the feasibility and advantages of using TRIZ in NSD have been verified in the empirical cases

under two different service contexts. In Case 1, the island cycling program is redesigned with suggested supplementary programs that can cater to the needs of other visitors who might not be interested in cycling while sightseeing. The entire problem is traced to an inherent contradiction in the island travel route design. The elimination of the contradiction provides us with many insightful solutions to address the problem. In Case 2, TRIZ tools reveal the inherent contradiction between customer dining needs and the canteen operation time. Based on recommendations from the 4 separation principles and 40 inventive principles, innovative service ideas such as contracting nighttime food outlet operators, establishing direct sales, and having an order system on campus were suggested. In both cases, ideas were generated throughout the entire service design process. In sum, we show that the TRIZ-based methodology is able to deliver a set of innovative solutions systematically.

In comparison with other service design tools, such as QFD and RCA, TRIZ is distinct with its unique nature in defining the problem and then generating and evaluating solutions. It can be seen in Table 4 that TRIZ-based service design methods illustrate a completely different new process of problem solving.

Managerial Implications

Several managerial implications accrue from this study. First, as shown in the case studies, the systematic method is able to help service developers to formulate effective solutions to address customer service problems. This method can replace the traditional ad hoc practices frequently used in developing new services in many service companies. Hence, service companies are likely to enhance their capacity to deliver innovative service offerings and shorten service development cycle and time to market.

Moreover, by having a powerful knowledge base that consists of a collection of innovation patterns, TRIZ can help practitioners develop new services by leveraging on previous innovation, thus avoiding reinventing the wheel. Extensive design experience may no longer be a necessary requirement for developing innovative services. With the implementation of TRIZ in service development, it is thus possible for less experienced service developers to think innovatively and generate creative service strategies. In fact, service organizations can always consider tailoring TRIZ tools and further enhancing the effectiveness of the knowledge base by benchmarking the best service practices across different industries.

TABLE 4
Comparison of TRIZ-Based Method With Quality Function Deployment and Root Cause Analysis in Service Design

<i>Service Design Process</i>	<i>TRIZ-Based Method</i>	<i>Quality Function Deployment</i>	<i>Root Cause Analysis</i>
Problem definition	Problem is defined in a structured manner. Data of problem situation are captured. Complicated problem is decomposed into a set of small, simple problems. Possible TRIZ tools: Innovative Situation Questionnaire; problem formulator; TOP function analysis; substance-field analysis; anticipatory failure analysis	House of quality can systematically identify problems and contradictory service attributes. However, it is unable to reveal the possible root cause of the problem.	Root cause analysis can help to identify the root cause of the problem. However, the information capture is not as systematic as the TRIZ process.
Problem resolution	Inherent contradiction of the problem is identified. Solutions are generated by eliminating the contradictions with TRIZ knowledge-based tools. Possible TRIZ tools: root contradiction analysis; separation principles; inventive principles; contradiction matrix; 76 standard solutions; ARIZ	No detailed step to guide service designers to proceed in contradiction resolution; solution generation relies on team members' experience and creativity.	No detailed step to guide service designers to proceed in root cause resolution; solution generation relies on team members' experience and creativity.
Solution evaluation	Generated ideas are prioritized with the closeness to the state of ideality. If possible, it is encouraged to try the implementation of the most ideal solution. Possible TRIZ tools: ideal final result.	No idea selection step/tool	No idea selection step/tool

NOTE: TRIZ = theory of inventive problem solving; TOP = tool-object-product; ARIZ = algorithm of inventive problem solving.

Limitations and Future Work

As the introduction of TRIZ to new service design is still in its infancy, there are several limitations that need to be addressed. The first is the need to concretize a TRIZ knowledge base in NSD. As TRIZ was first conceived and developed in engineering, many of its principles and tools were originally designed to resolve technical problems instead of service problems. Future research can focus on identifying patterns of service innovations and incorporating more TRIZ tools, laws, and principles in service problem solving. A service product is different from a physical product because of its distinct characteristics: intangibility, heterogeneity, simultaneity, customer participation, and perishability. This difference might indicate a unique track of evolution when it comes to patterns of service innovation. Therefore, to capture the nature of service innovation, a more comprehensive TRIZ and its knowledge base needs to be developed.

Another limitation of this study is that only two case studies were conducted. To overcome this limitation, future research can further validate the feasibility of using the TRIZ method in solving various kinds of service problems. The objective is to enable TRIZ to resolve a much wider range of service problems (e.g., problems with different service design attributes and in different service industries). To achieve this goal, more empirical case studies need to be carried out in different service industries and service contexts.

CONCLUSION

This article proposes a new approach to service design based on the TRIZ method. Several TRIZ problem-solving tools are modified and embedded into the new model. With the incorporation of TRIZ systematic thinking, this approach realizes the aim of systematic design and innovation in the service domain. The findings obtained from case studies suggest that the process of idea generation can be better planned and controlled with formalized operation procedures. It also indicates that this type of formalized mechanism can be extended to other phases of service development, such as business analysis and market testing. TRIZ may have been created on the basis of technical as opposed to nontechnical data. Nonetheless, its potential to unleash innovativeness in the service industries holds great promise.

APPENDIX A

Forty Inventive Principles With Examples in Services¹

Principle 1. Segmentation

- Divide an object or system into independent parts.
- Make an object or system easy to disassemble.
- Increase the degree of fragmentation or segmentation.

1. The examples listed are only approximate illustrations. Some principles (e.g., Principles 14, 37, and 38) are so mechanically oriented that a corresponding service example is not possible or appropriate.

Examples

- Consumers can be segmented based on their needs, age group, and buying behavior.
- In most service industries, the service package is a mix of tangible and intangible elements (e.g., the atmosphere of a restaurant and the friendliness of its waiters might be as important as the taste of the food it sells).

Principle 2. Extraction

- Separate an interfering part or property from an object or system, or single out the only necessary part (or property) of an object or system.

Examples

- Banks set up e-portals to provide services such as fund transfer, bill payment, and simple advisory, allowing branches to focus on primarily cash-based transactions (i.e., cash withdrawals and deposits) that cannot be done over Internet.
- Cinemas set up automated ticket booking machines at residential areas or provide phone booking services so that consumers do not have to travel to the cinema to buy tickets.

Principle 3. Local quality

- Change an object's or system's structure from uniform to nonuniform; change an external environment (or external influence) from uniform to nonuniform.
- Make each part of an object or system function in conditions most suitable for its operation.
- Make each part of an object or system fulfill a different and useful function.

Examples

- Service offerings can be customized based on the needs of consumers. For instance, DELL provides flexibility for computer configuration.
- The layout design in large grocery stores, such as Wal-Mart, emphasizes strategic product placement and consumer flows through their stores to maximize sales and convenience (Metters, King-Metters, and Pullman 2003).

Principle 4. Asymmetry

- Change the shape of an object or system from symmetrical to asymmetrical.
- If an object or system is asymmetrical, increase its degree of asymmetry.

Example

- Sometimes, providing personalized services instead of standardized services can help to create a unique experience to consumers. For instance, consumers are greeted by their names at hotel reception counters, and hair salons make records of consumer preferences.

Principle 5. Consolidation

- Bring closer together (or merge) identical or similar objects or systems; assemble identical or similar parts to perform parallel operations.
- Make operations contiguous or parallel; bring them together in time.

Example

- Bundling different service offerings and operating them together. For instance, a combined admission ticket to an amusement park means visitors can enjoy a variety of attractions, which helps to create lasting impressions.

Principle 6. Universality

- Make a part, an object, or system perform multiple functions; eliminate the need for other parts.
- Use standardized features.

Examples

- Service can be designed in a way to meet various customer needs. A customer service officer at a shopping mall can function as a telephone operator as well as a customer service officer.
- Franchise food outlets, such as McDonald's and Starbucks, deliver standard food dishes and service facilities to consumers.

Principle 7. Nesting

- Place one object or system inside another; place each object or system, in turn, inside the other.
- Make one part pass through a cavity in the other.

Example

- Long-distance flight can be a very boring and tiring experience if no additional service or "entertainment," such as games or movies, is provided during the flight.

Principle 8. Counterweight

- To compensate for the weight of an object or system, merge it with other objects or systems that provide lift.
- To compensate for the weight of an object or system, make it interact with the environment.

Example

- To cope with tremendous marketing expense, newly emerged e-banks often ally with established traditional banks to gain rapid recognition from the market.

Principle 9. Prior counteraction

- If it is necessary to do an action with both harmful and useful effects, this action should be replaced with anti-actions to control the harmful effects.
- Create beforehand stresses in an object or system that will oppose known undesirable working stresses later on.

Examples

- Companies provide a money-back guarantee or extended after-sale service for new products.
- Companies educate customers about potential fallout of their products (e.g., displaying disclaimers to avoid a potential liability claim).

Principle 10. Prior action

- Perform, before it is needed, the required change of an object or system (either fully or partially).

- Prearrange objects or systems such that they can come into action from the most convenient place and without losing time for their delivery.

Examples

- Multinational companies offer international warranty services for customers.
- To shorten the checkout time, many hotels total the bills and slide them under the guest room doors during the last night of their stays, thereby achieving "minimum waiting time."

Principle 11. Cushion in advance

- Prepare emergency means beforehand to compensate for the relatively low reliability of an object or system.

Example

- To balance between limited capacity and uneven consumer demands, service firms can use a set of preventive strategies such as price differentiation to encourage demand during off-peak times and appointment-only reservations to control service quality.

Principle 12. Equipotentiality

- In a potential field, limit position changes (e.g., change operating conditions to eliminate the need to raise or lower objects or systems in a gravity field).

Example

- By establishing a network of branches, car rental companies enable customers to rent cars in one branch and return them to another branch.

Principle 13. Do it in reverse

- Invert the action(s) used to solve the problem (e.g., instead of cooling an object or system, heat it).
- Make movable parts (or the external environment) fixed and fixed parts movable.
- Turn the object (or process) "upside down."

Example

- Provide a "mobile food stall" to remote industrial areas where workers might need to travel far for food.

Principle 14. Spheroidality

- Instead of using rectilinear parts, surfaces, or forms, use curvilinear ones; move from flat surfaces to spherical ones, from parts shaped as a cube (parallelepiped) to ball-shaped structures.
- Use rollers, balls, spirals, and domes.
- Go from linear to rotary motion; use centrifugal forces.

Principle 15. Dynamicity

- Allow (or design) the characteristics of an object, external environment, or process to change to be optimal or find an optimal operating condition.
- Divide an object or system into parts capable of movement relative to each other.
- If an object (or process) is rigid or inflexible, make it movable or adaptive.

Example

- Service firms can empower frontline staff to have discretionary power in delivering services such as giving discounts if customers order a large amount of the service.

Principle 16. Partial or excessive action

- If 100% of an object or system is hard to achieve using a given solution method, then, by using "slightly less" or "slightly more" of the same method, the problem may be considerably easier to solve.

Example

- Giving advance notice and explanation to customers for temporary unavailability of a service can prevent loss of consumer loyalty due to waiting. For instance, Web sites can put up notices such as "server upgrading" to explain the temporary shutdown or slow response rate.

Principle 17. Transition into a new dimension

- Move an object or system in two- or three-dimensional space.
- Use a multistory arrangement of objects or systems instead of a single-story arrangement.
- Tilt or reorient the object or system; lay it on its side.
- Use "another side" of a given area.

Example

- The organizational structure of McDonald's is pyramid shaped, with layers of supervision from the assistant store manager, store manager, and regional manager to corporate "consultants," ensuring consistency of service delivery across all locations (Fitzsimmons and Fitzsimmons 2001).

Principle 18. Mechanical vibration

- Cause an object or system to oscillate or vibrate.
- Increase its frequency (even up to the ultrasonic).
- Use an object's or system's resonant frequency.
- Use piezoelectric vibrators instead of mechanical ones.
- Use combined ultrasonic and electromagnetic field oscillations.

Example

- Airlines use a ticket pricing system to maximize profit from vibrating demands.

Principle 19. Periodic action

- Instead of continuous action, use periodic or pulsating action.
- If an action is already periodic, change the periodic magnitude or frequency.
- Use pauses between impulses to perform a different action.

Examples

- Collecting customer feedback on a regular basis can help service organizations to monitor customer satisfaction.
- Consumer demand in some service industries exhibits very cyclical behavior over different periods of time, with considerable variation between the peaks and valleys. Ser-

vice firms can profit from this characteristic by offering additional services. For instance, airlines open more flights to popular destinations during peak season.

Principle 20. Continuity of useful action

- Carry on work continuously; make all parts of an object or system work at full load all the time.
- Eliminate all idle or intermittent actions or work.

Examples

- Universities usually maintain good relationships with their alumni, even long after students have left their alma mater. This allows the university to raise donations as well as to sell continuing/executive education to their alumni.
- Management consulting firms typically maintain close relationships with their ex-employees, who are likely to help to bring in new business.

Principle 21. Rushing through

- Conduct a process or certain stages (e.g., destructible, harmful, or hazardous operations) at high speed.

Example

- Keeping customers in a long queue risks losing their loyalty. One of the possible means to overcome this is to let customers feel that waiting time was skipped psychologically by keeping them occupied with entertainment (e.g., Disney employees entertaining customers waiting in queue) or distraction (high-rise buildings put mirrors on lift doors to make people less maniacal during waits).

Principle 22. Convert harm into benefit

- Use harmful factors (particularly harmful effects of the environment or surroundings) to achieve a positive effect.
- Eliminate the primary harmful action by adding it to another harmful action to resolve the problem; amplify a harmful factor to such a degree that it is no longer harmful.

Examples

- Safety or other forms of audits can be used to identify performance gaps even though such audits often take up a great amount of resources.
- In professional services such as management consulting, the price of a job is often considered as a surrogate for service quality. Thus, increasing the price may bring a high-quality perception among the customers.

Principle 23. Feedback

- Introduce feedback (referring back, cross-checking) to improve a process or action.
- If feedback is already used, change its magnitude or influence.

Example

- Keeping consumer records and listening to their feedback will be helpful for service firms to improve service excellence.

Principle 24. Mediator

- Use an intermediary carrier article or intermediary process.

- Merge one object temporarily with another (which can be easily removed).

Example

- Service companies may choose to outsource noncore functions or functions that lack economies of scale to other companies that essentially merge similar functions from different companies.

Principle 25. Self-service

- Make an object or system serve itself by performing auxiliary helpful functions.
- Use waste resources, energy, or substances.

Example

- Customer participation in service delivery can be an interesting experience or even an attraction. Many zoos allow visitors to feed certain animals or participate in some animal shows.

Principle 26. Copying

- Instead of an unavailable, expensive, fragile object or system, use simpler and inexpensive copies.
- Replace an object or system or process with optical copies.
- If visible optical copies are already used, move to infrared or ultraviolet copies.
- Copy creative service concepts across different industries.

Example

- The concept of the automatic vending machine has been copied and applied across different industries to provide a wide range of services such as returning library books, posting letters, and dispensing cash.

Principle 27. Dispose

- Replace an expensive object or system with multiple inexpensive objects or systems, compromising certain qualities (such as service life).

Examples

- School textbooks may come with software with limited functionality or limited copyright, which expires after one or two semesters.
- Trial versions of software can often be downloaded for users to try out before making a decision to buy the complete version.

Principle 28. Replacement of mechanical system

- Replace a mechanical means with a sensory (optical, acoustic, taste, or smell) means.
- Use electric, magnetic, and electromagnetic fields to interact with the object or system.
- Change from static to movable fields, from unstructured fields to those having structure.
- Use fields in conjunction with field-activated (e.g., ferro-magnetic) particles.

Examples

- Videotapes of lectures and CD recordings of concerts represent convenient substitutes for physical attendance.

- The development of e-services eliminates the necessity for physical traveling (e.g., videoconference, Web-based class, online order, and payment).

Principle 29. Pneumatic or hydraulic construction

- Use intangible parts of an object or system instead of tangible parts.

Example

- Companies may provide free services to a charity organization to raise their profile instead of spending money on advertisement.

Principle 30. Flexible membranes or thin films

- Use flexible shells and thin films instead of three-dimensional structures.
- Isolate the object or system from the external environment using flexible shells and thin films.

Example

- Restaurants may separate smoking and nonsmoking sections through flexible partitions and air-exhaust systems, depending on the demand.

Principle 31. Porous material

- Make an object or system porous or add porous elements (inserts, coatings, etc.).
- If an object or system is already porous, use the pores to introduce a useful substance or function.

Example

- Customs areas at airports or border checkpoints have a green lane if visitors have nothing to declare, in contrast to the red lane if visitors have brought items that need to be declared.

Principle 32. Changing the color

- Change the color of an object or system or its external environment.
- Change the transparency of an object or system or its external environment.

Example

- Changing the color of a service facility might influence customers' perception of the service (e.g., renovate restaurants with warm colors in winter; use a hospital color "association," which might be helpful for rapid recovery for patients).

Principle 33. Homogeneity

- Make objects or systems interact with a given object or system of the same material (or material with identical properties).

Example

- Some hospitals encourage patients who have recovered from illnesses to discuss their experiences with new patients to alleviate their preoperative fears about certain types of treatment.

Principle 34. Rejecting and regenerating parts

- Make portions of an object or system that have fulfilled their functions go away (discard by dissolving, evaporating, etc.) or modify these directly during operation.
- Conversely, restore consumable parts of an object or system directly in operation.

Example

- Amenities such as food stalls and temporary toilets can be dissolved after the carnival is over.

Principle 35. Transformation of properties

- Change an object's or system's physical state (e.g., to a gas, liquid, or solid).
- Change the concentration or consistency.
- Change the degree of flexibility.
- Change the atmosphere to an optimal setting.

Example

- Hotels may offer a simplified check-in or free room upgrade for regular guests.

Principle 36. Phase transition

- Use phenomena occurring during phase transition (e.g., volume changes, loss or absorption of heat, etc.).

Examples

- As members grow older, some resort clubs may provide more family or retiree-oriented activities to cater for the evolution of members' needs.
- Recreation centers launch new leisure programs in different seasons.

Principle 37. Thermal expansion

- Use thermal expansion (or contraction) of materials.
- If thermal expansion is being used, use multiple materials with different coefficients of thermal expansion.

Principle 38. Accelerated oxidation

- Replace common air with oxygen-enriched air.
- Replace enriched air with pure oxygen.
- Expose air or oxygen to ionizing radiation.
- Use ionized oxygen.
- Replace ozonized (or ionized) oxygen with ozone.

Principle 39. Inert environment

- Replace a normal environment with an inert one.
- Add neutral parts or inert additives to an object or system.

Principle 40. Composite materials

- Change from uniform to composite (multiple) materials.

Example

- Mix a service with other services or tangible elements. For instance, airlines give away souvenirs to passengers, and hotels provide complimentary toiletry items with the hotel name prominently affixed.
-

APPENDIX B

Four Separation Principles With Examples in Service Development

1. Separation in Space

This principle is to separate the contradictory requirements or functions in a system into different subsystems, so that these subsystems can perform their own functions without contradicting each other.

Example: Hospitals are often separated into different departments into different space zones. Based on their own needs, patients can approach the relevant departments for consultation without disturbing other departments.

2. Separation in Time

This principle is to separate the contradictory requirements in a system into different timelines, so that these requirements or functions can be met or operate at different time schedules without contradicting each other.

Example: Entertainment places such as cinemas, restaurants, and resorts set different pricings (e.g., daytime discounts, early-bird discount) for different periods to balance customer demands.

3. Separation Between the Whole and Its Parts

This principle is to separate the contradictory requirements or functions between the whole system and its subsystems.

Example: Areas allowing smoking are strictly separated from nonsmoking areas in many places, such as in trains, restaurants, and so on. Thus, customers with differing needs can all be served.

4. Separation Upon Conditions

This principle is to separate contradictory requirements in a condition by changing the setting. It may be helpful to stimulate the production of useful functions and suppress the production of harmful functions.

Example: Some restaurants provide house drinks to customers who have to wait for a long period of time (i.e., harmful function) for their food. This reinforces the impression of good food quality (i.e., useful function) at the restaurant.

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